

Hisu Kim

tomm1203@gm.gist.ac.kr | [Google Scholar](#) | hiskim1.github.io

EDUCATION

Gwangju Institute of Science and Technology, South Korea <i>M.S. in Climate Analysis and Modeling Lab (Advisor: Jin-ho Yoon)</i>	Sep. 2025 – Aug. 2027 (expected) TGPA 4.37/4.5
Gwangju Institute of Science and Technology, South Korea <i>B.S. in Electrical Engineering and Computer Science and Earth Science and Environmental Engineering Minor in Mathematics</i>	2019 – 2025 TGPA 4.10/4.5
Warsaw University of Technology, Poland <i>Exchange student, Faculty of Building Services, Hydro and Environmental Engineering</i>	Oct. 2022 – Feb. 2023 TGPA 4.86/5.0
University of California, Berkeley, USA <i>Summer Session C</i>	June – Aug. 2023 TGPA 4.0/4.0

PUBLICATIONS AND CONFERENCES

Publications (*Equal contribution)

- Cho, G. *, **Kim, H. ***, & Lee, S. (2026). *Literature designed in the language of science: Yi Sang's "Memorandum on the Line" and the paradigm shift* [Manuscript in preparation].
- **Kim, H. ***, Jang, W. *, Ryu, J. *, Kwon, M., Son, S.-W., Jeong, J.-H., Kim, H., & Yoon, J.-H. (2026). *Tracing uncertainty in deep-learning post-processing of subseasonal forecasts* [Manuscript in preparation].
- **Kim, H.**, Ryu, J., Son, S.-W., Jeong, J.-H., Kim, H., & Yoon, J.-H. (2026). A spectral test of the butterfly effect and physical consistency in the diffusion-based GenCast's ensembles. *npj Climate and Atmospheric Science*, 9, Article 110. <https://doi.org/10.1038/s41612-026-01380-1>
- Ryu, J., **Kim, H.**, Wang, S.-Y., & Yoon, J.-H. (2026). Increasing resolution and accuracy in sub-seasonal forecasting through 3D U-Net: The western US. *Geoscientific Model Development*, 19(1), 27–39. <https://doi.org/10.5194/gmd-19-27-2026>

Conferences

- **Kim, H.**, Ryu, J., & Yoon, J.-H. (2026, December 7–11). *What's going on inside GraphCast? Diagnosing GraphCast's latent graph signals behind blurry forecast* [Oral presentation]. AGU26 Annual Meeting, San Francisco, CA, United States.
- **Kim, H.**, & Yoon, J.-H. (2026, October 14–16). *그래프 푸리에 변환과 디리클레 에너지를 이용한 GraphCast 잠재공간의 예보 평활화 진단* [Diagnosing forecast smoothing in GraphCast's latent space using the graph Fourier transform and Dirichlet energy] [Oral presentation]. Korean Meteorological Society Fall Meeting 2026, Gyeongju, South Korea.
- **Kim, H.**, Ryu, J., Son, S.-W., Jeong, J.-H., Kim, H., & Yoon, J.-H. (2026, October 14–16). *인공지능 확률예보모델에서 나타난 중규모 오차성장 부재와 운동에너지 축적* [Absence of mesoscale error growth and kinetic energy accumulation in the AI probabilistic forecast model GenCast] [Oral presentation]. Korean Meteorological Society Fall Meeting 2026, Gyeongju, South Korea.
- **Kim, H.**, Ryu, J., & Yoon, J.-H. (2026, October 11–14). *Tracing blurry forecast inside GraphCast: A latent space diagnosis via spectral graph theory* [Oral presentation]. Asian Conference on Meteorology 2026, Gyeongju, South Korea.
- **Kim, H.**, Ryu, J., & Yoon, J.-H. (2026, October 11–14). *Spectral diagnosis of GenCast's jet stream kinetic energy: Noise artifacts in probabilistic AI weather models* [Poster presentation]. Asian Conference on Meteorology 2026, Gyeongju, South Korea.
- **Kim, H.**, Ryu, J., Jang, W., Kwon, M., & Yoon, J.-H. (2026, September 7–11). *Uncertainty assessment in ensemble forecast post-processing using U-Net and diffusion* [Poster presentation]. Third International Conference on Subseasonal to Seasonal to Decadal Prediction (S2S2D), Reading, United Kingdom.
- **Kim, H.**, & Yoon, J.-H. (2025, December 15–19). *Revealing the butterfly effect in deep learning weather prediction model via kinetic energy spectra change* [Oral presentation]. AGU25 Annual Meeting, New Orleans, LA, United States.

PROFESSIONAL DEVELOPMENT

Rosspalooza 2026 <i>University of Chicago, IL, USA</i>	20 – 31 July 2026
Presented “40 vs 10^{30} years? How much data is needed to predict the weather?”	
Advanced School on Foundation Models for Scientific Discovery <i>ITCP, Italy (virtual)</i>	10 – 18 July 2025
Summer School of Mathematics: Algorithms <i>Eötvös Loránd University, Hungary</i>	23 – 27 June 2025

SCHOLARSHIPS AND AWARDS

2026 Weather Big Data Contest <i>Korea Meteorological Administration</i>	Aug. 2026
Grand Prize, Minister of Climate, Energy and Environment Award	
Presidential Science Scholarship for Graduate Student <i>Ministry of Science and ICT</i>	2026-7 (1-year)
Government Funded Scholarship for GIST Student <i>GIST</i>	2019 Spring – Present
Scholarship for Outbound Program <i>GIST</i>	2023 Fall
Academic Excellence Scholarship <i>GIST</i>	4 Semesters
Scholarship for Summer Session Program Abroad <i>GIST</i>	2023 Summer

TEACHING EXPERIENCE

Calculus (Single- & Multivariable) <i>Teaching Assistant</i>	2025 Spring – 2026 Fall
- Single-variable (2026 Spring): designed weekly problem sets, prepared and led recitation sessions, and provided written feedback on students’ solutions - Multivariable (2025 Spring & Fall, 2026 Fall): held weekly office hours, answering conceptual questions and guiding students through theorem proofs	
Graph Theory <i>Teaching Assistant</i>	2024 Fall
- Conducted weekly office hours, answering questions and guiding students through theorem proofs - Organized regular sessions to provide detailed feedback on homework and quizzes, and advanced mathematical $\text{\LaTeX}$ usage - Offered supplementary lessons on challenging topics and extra problems	
Literature Courses from Prof. Soo-Jeong Lee <i>Teaching Assistant</i>	2023 Spring – 2026 Spring
- Built syllabus generator website to streamline the professor’s course preparation process - Used $\text{\LaTeX}$ to edit e-books compiled from students’ story-writing assignments	
Digital Design <i>Teaching Assistant</i>	2023 Spring
- Conducted office hours twice a week answering questions from homework - Created sample exam questions to facilitate student learning	

WORK EXPERIENCE

Cybersecurity Analyst	Feb. 2021 – Oct. 2022
<i>Republic of Korea Naval Fleet Command</i>	<i>Computer Emergency Response Team</i>
- Co-administered dozens of Linux production servers across the Fleet Command network - Mitigated <i>log4j</i> and <i>MS-MSDT</i> vulnerabilities via NAC-distributed registry patches and removal of exploitable files; analyzed firewall/IPS packets and logs to flag unauthorized or anomalous communications	

SKILLS

Programming	Python	(Main)
	C/C++, C#, JavaScript, Bash	(Intermediate)
Libraries	PyTorch (Lightning), JAX, xarray, Dask, zarr	
Tools	Linux, Git, $\text{\LaTeX}$, NCL, GrADS	
Language	Korean (native)	
	English (fluent)	
	Spanish, Polish (elementary)	